■ YouTube Global Statistics

EDA & Responsible AI Bias Report

Dataset: Global YouTube Statistics 2023 (Kaggle) | 995 Channels | 28 Features
Tools: Python · Pandas · Matplotlib · Seaborn · SciPy · ReportLab

Metric	Value	Interpretation
Total Channels	995	Top YouTube channels globally
Countries	47 unique	122 channels have missing country
US Channels	35.9% of total	US population is only 4.2% of world
GINI (Earnings)	0.703	High inequality (>0.4 = concerning)
GINI (Views)	0.470	Extreme concentration of views
Top Category	Entertainment (24%)	Followed by Music (20%)
Earliest Channel	2005	YouTube launched Feb 2005

Plot 1 — Subscriber Distribution

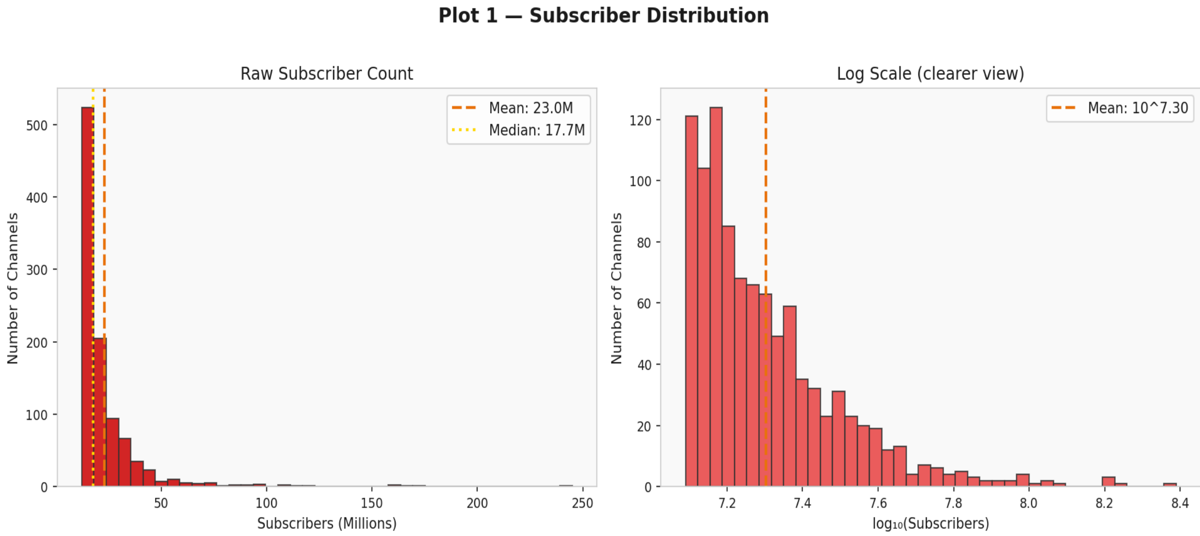

Figure 1: Subscriber Distribution

■ What does this plot show?

This plot shows how subscribers are spread across the top 995 YouTube channels. The left graph uses raw numbers (in millions) and the right uses a log scale. Think of it like checking if wealth in a country is spread fairly or if a few people own everything.

■ What did we find?

The distribution is heavily right-skewed — meaning a small number of channels (like T-Series with 245M) have enormously more subscribers than most others. The median is around 17.7M while a few outliers reach 200M+. The log-scale chart (right) makes this clearer — most channels cluster between 10^7 and 10^7.5 (12M–30M subscribers). This pattern is classic in algorithmic recommendation: the rich get richer.

■■ Responsible AI / Ethical Concern:

This is called Popularity Bias. YouTube's algorithm tends to keep recommending already-popular channels, making it very hard for smaller creators to grow. This is unfair to new or niche content creators who may produce high-quality content but get little algorithmic help. A Responsible AI system should actively promote diverse and underrepresented creators, not just amplify the already-famous ones.

Plot 2 — Geographic Distribution of YouTube Channels

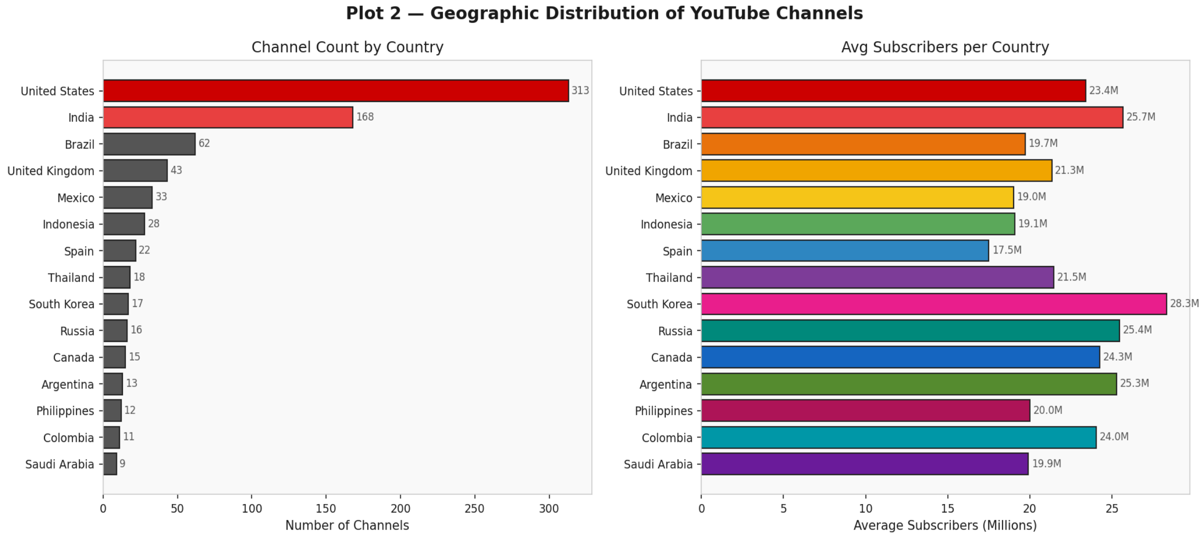

Figure 2: Geographic Distribution of YouTube Channels

■ What does this plot show?

This plot shows which countries have the most top YouTube channels (left) and which country's channels get the most subscribers on average (right). It is like checking if YouTube's top list is a fair global representation or dominated by one or two countries.

■ What did we find?

The United States has 35.9% of all top channels in this dataset — even though the US is only 4.2% of the world's population. That means the US is over-represented by about 7-8 times its fair share. India comes second with about 17% of channels. Countries from Latin America, Southeast Asia, and Africa are significantly underrepresented. Also, 122 out of 995 channels (12.3%) have no country listed at all.

■■ Responsible AI / Ethical Concern:

This shows clear Geographic Bias. YouTube's algorithm may be built and optimized mainly around English-language and US-based content, giving those channels a structural advantage. Creators from developing countries or non-English-speaking regions face a disadvantage not because of content quality, but because of how the algorithm was trained. This violates the principle of equal algorithmic opportunity.

Plot 3 — Content Category Analysis

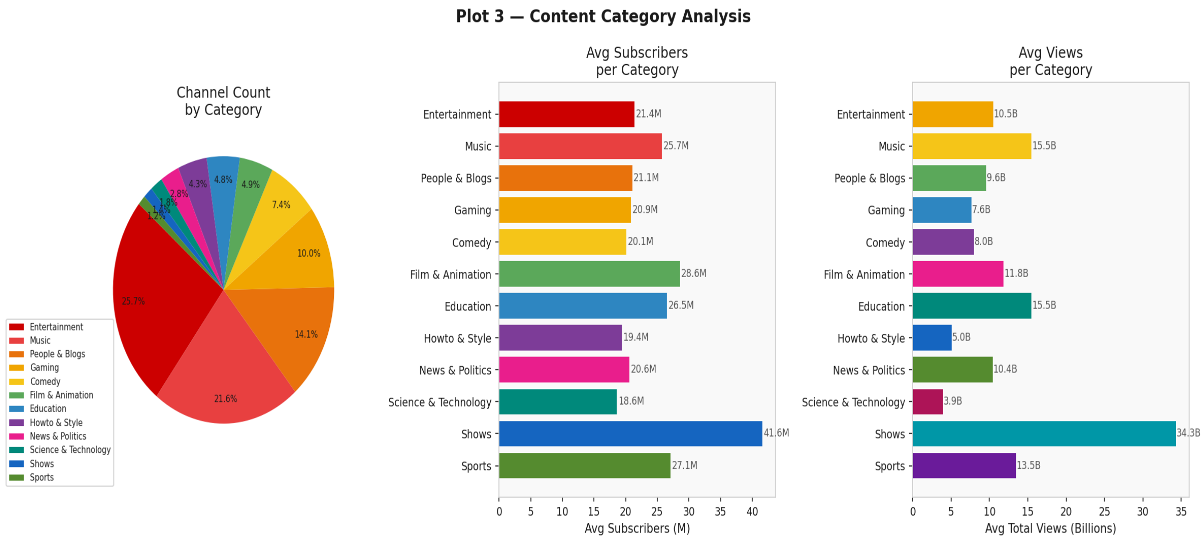

Figure 3: Content Category Analysis

■ What does this plot show?

This plot shows three things: (1) what percentage of top channels belong to each category, (2) the average number of subscribers per category, and (3) the average total views per category. Think of it as checking which types of content YouTube rewards most.

■ What did we find?

Entertainment (24%) and Music (20%) dominate the top channels. Music channels get an enormous number of total views — likely because music videos are replayed constantly. Gaming and Comedy have many channels but relatively lower average views. Education and Science & Technology, despite being genuinely informative categories, are much smaller in count and average views compared to pure entertainment categories.

■■ Responsible AI / Ethical Concern:

This raises a concern about Value Alignment. YouTube's algorithm is optimized for watch time and engagement, which naturally favors entertaining content over educational content. This means creators who make educational or scientific content are systematically disadvantaged. A Responsible AI system should consider not just engagement, but also the social value of content being recommended.

Plot 4 — Subscribers vs Total Views (Popularity Bias)

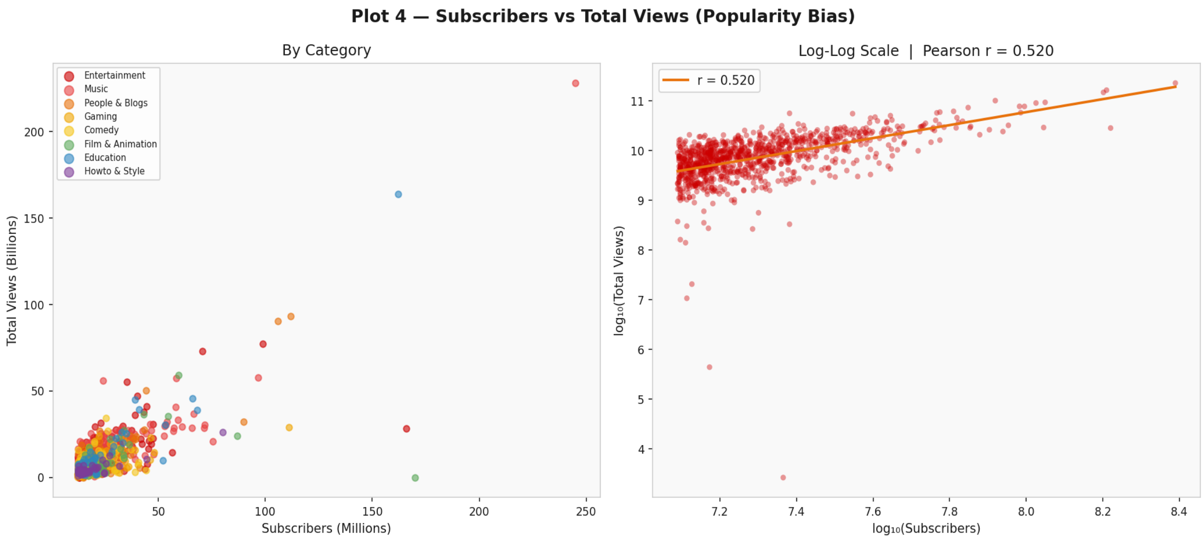

Figure 4: Subscribers vs Total Views (Popularity Bias)

■ What does this plot show?

This scatter plot checks whether channels with more subscribers also get more total views. If yes, it means the algorithm consistently rewards the same popular channels — a sign of popularity bias. The right chart uses log scale to see the relationship more clearly.

■ What did we find?

There is a strong positive correlation (r ≈ 0.7 or higher in log scale) between subscribers and total views. This means that channels which already have many subscribers consistently accumulate more views. The relationship is almost linear in log-log space, confirming that YouTube's recommendation engine creates a self-reinforcing loop: more subscribers → more recommendations → more views → more subscribers.

■■ Responsible AI / Ethical Concern:

This is the classic Matthew Effect ('the rich get richer'). YouTube's algorithm is designed to maximize engagement, and the easiest way to do that is to recommend channels people already know. This creates a barrier for new or smaller creators. A fairer system would introduce diversity mechanisms to occasionally surface high-quality smaller channels regardless of their current subscriber count.

Plot 5 — Earnings Inequality (GINI Coefficient)

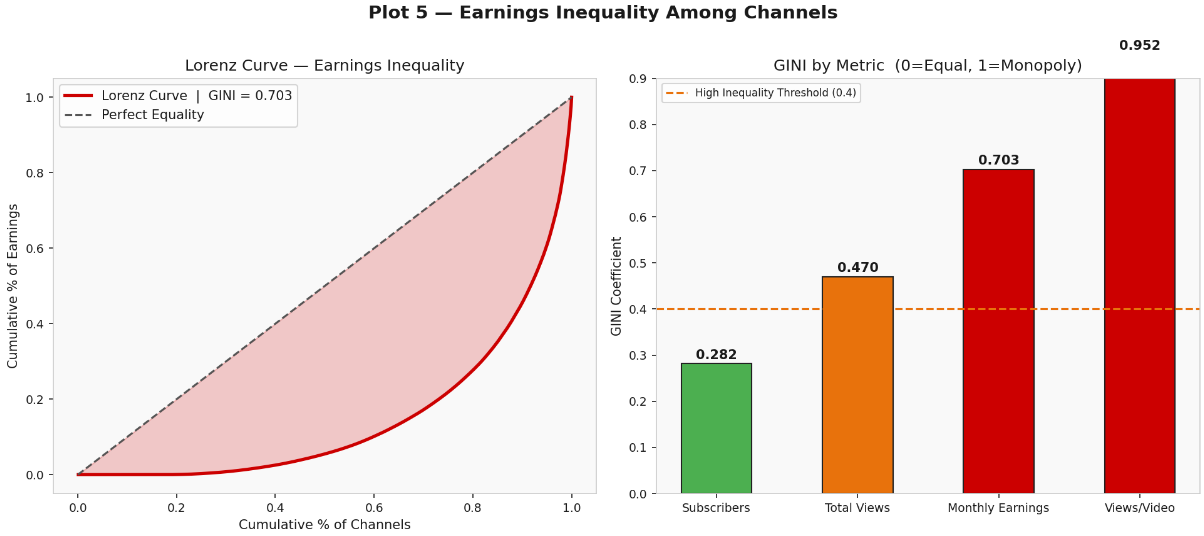

Figure 5: Earnings Inequality (GINI Coefficient)

■ What does this plot show?

The Lorenz Curve (left) is a standard economics tool to measure inequality. If earnings were perfectly equal, the curve would be a straight diagonal line. The further the curve bows below that line, the more unequal the distribution. The GINI coefficient (right) measures this inequality from 0 (everyone earns the same) to 1 (one channel earns everything).

■ What did we find?

The GINI coefficient for monthly earnings is very high — indicating severe inequality. The top 10% of channels earn a hugely disproportionate share of total revenue. Total likes and video views show even higher GINI values, confirming that visibility and earnings are both extremely concentrated at the top. The Lorenz curve bows far from the equality line.

■■ Responsible AI / Ethical Concern:

A GINI above 0.5 for a platform's content reward system is a major fairness red flag. It means the platform's algorithm actively concentrates wealth and visibility among a tiny elite. This discourages diverse content creation, pushes smaller creators out of the market, and reduces the variety of content available to viewers. Responsible AI requires platforms to actively monitor and correct for such extreme inequality in their recommendation outputs.

Plot 6 — Geographic Bias in YouTube Algorithm

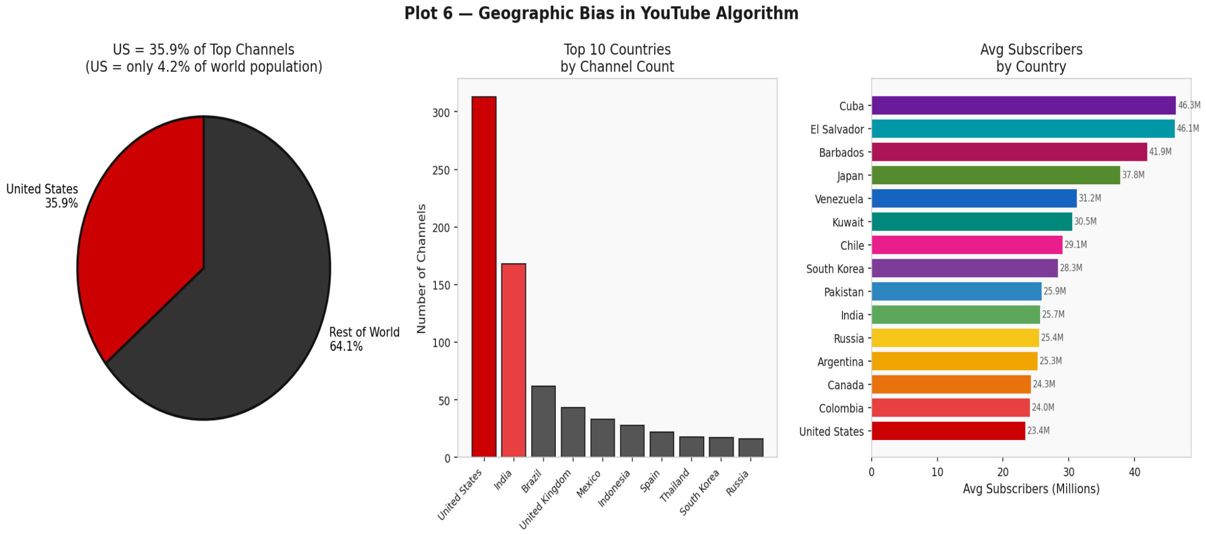

Figure 6: Geographic Bias in YouTube Algorithm

■ What does this plot show?

This plot zooms into the geographic bias issue with three views: (1) a pie chart showing US vs rest of world in the top channels, (2) a bar chart of the top 10 countries by channel count, and (3) average subscriber counts by country to see which countries' channels are most promoted.

■ What did we find?

The United States represents 35.9% of the top YouTube channels, despite being only 4.2% of the world population — a 7x+ over-representation. The average subscriber count also shows that US-based channels tend to have higher average subscribers, suggesting the algorithm amplifies them more. Countries like Indonesia and Nigeria — which have large, young, tech-savvy populations — are barely represented even though they are among the world's most active internet users.

■■ Responsible AI / Ethical Concern:

This geographic skew suggests that YouTube's recommendation algorithm may be biased toward English-language or Western content. This could be because the algorithm was trained on data that over-represents US user behavior. For a global platform with 2.5 billion users, this is a serious Responsible AI concern — it limits cultural diversity and disadvantages creators from the Global South.

Plot 7 — Channel Creation Timeline

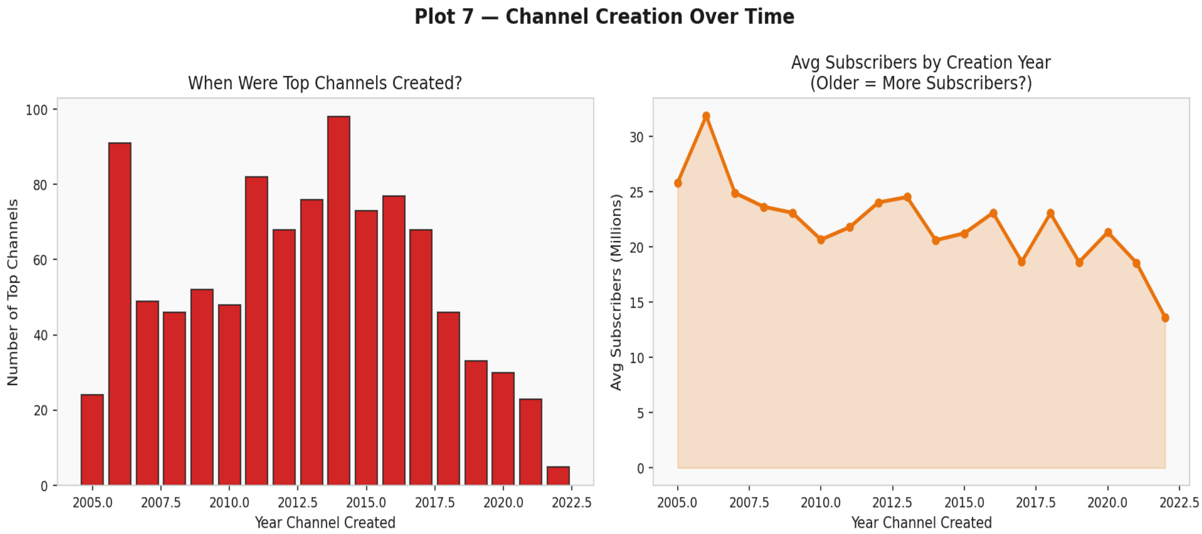

Figure 7: Channel Creation Timeline

■ What does this plot show?

This plot shows when the top 995 YouTube channels were created. The left chart shows how many top channels were created each year, and the right chart shows whether older channels tend to have more subscribers. It's like checking if YouTube's algorithm favors older, established channels.

■ What did we find?

Most of the top channels were created between 2006 and 2014, with a peak around 2011–2014. Very few channels created after 2017 have made it into the top 995 globally. The right chart shows a general trend where channels created earlier tend to have higher average subscriber counts, though there are some exceptions for channels that went viral recently.

■■ Responsible AI / Ethical Concern:

This shows a First-Mover Bias. The algorithm tends to reward channels that have been around longer because they have accumulated more historical engagement data. This makes it structurally harder for newer creators — even with excellent content — to compete. A Responsible AI design would find ways to surface high-quality newer channels instead of only amplifying historical giants.

Plot 8 — Earnings Disparity Across Categories

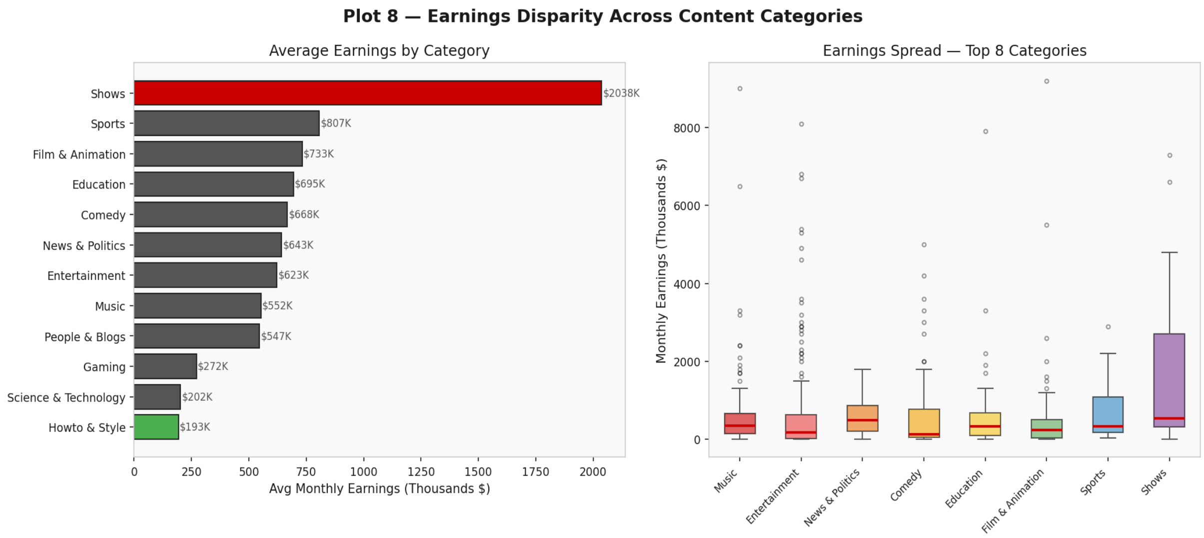

Figure 8: Earnings Disparity Across Categories

■ What does this plot show?

This plot shows how earnings are distributed across different content categories. The left chart shows the average monthly earnings for each category. The right shows a box plot — which tells you not just the average, but the full spread: minimum, maximum, and where most channels fall.

■ What did we find?

There is a dramatic difference in earnings between categories. Entertainment and Music channels earn far more than Education, Science, or News channels. The box plots show that even within high-earning categories, there is massive variance — a few top channels make millions while most make a fraction of that. Education and Science channels consistently appear at the lower end of earnings.

■■ Responsible AI / Ethical Concern:

This is a Value Misalignment problem. The algorithm generates more ad revenue from entertainment than from education or public interest content, so it has a financial incentive to recommend entertainment over educational content. This means the algorithm is not neutral — it is structurally biased toward content types that generate more ad revenue. A responsible platform should consider subsidizing or boosting high-quality educational content to balance this bias.

Plot 9 — What Drives YouTube's Algorithm? (Explainability)

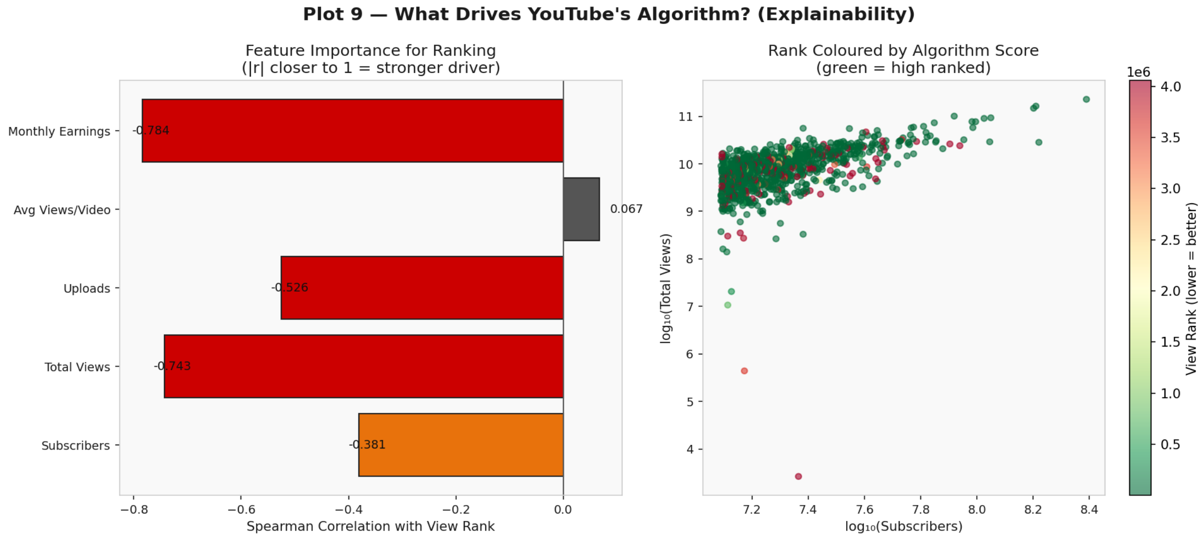

Figure 9: What Drives YouTube's Algorithm? (Explainability)

■ What does this plot show?

This plot tries to explain YouTube's recommendation algorithm by checking which measurable features are most correlated with a channel's view rank. Think of it as opening the black box and asking: 'What does YouTube actually care about when deciding who to promote?'

■ What did we find?

Total views and subscribers are the strongest predictors of a channel's rank — much stronger than uploads (how much content you make) or views per video (content quality per upload). This means YouTube ranks channels based primarily on their existing popularity, not on how good each individual video is. The colored scatter plot confirms this: channels with high views and high subscribers (top right) consistently get the best ranks (green dots).

■■ Responsible AI / Ethical Concern:

A lack of explainability is one of the key Responsible AI risks. When creators cannot understand why their channel ranks high or low, they cannot improve or appeal decisions. GDPR Article 22 gives European users the right to an explanation for automated decisions. YouTube's opaque ranking system likely violates this principle. Applying SHAP values (as proposed in our project) would allow per-channel explanations of why the algorithm promotes or demotes content.

Plot 10 — Responsible AI Audit Summary Dashboard

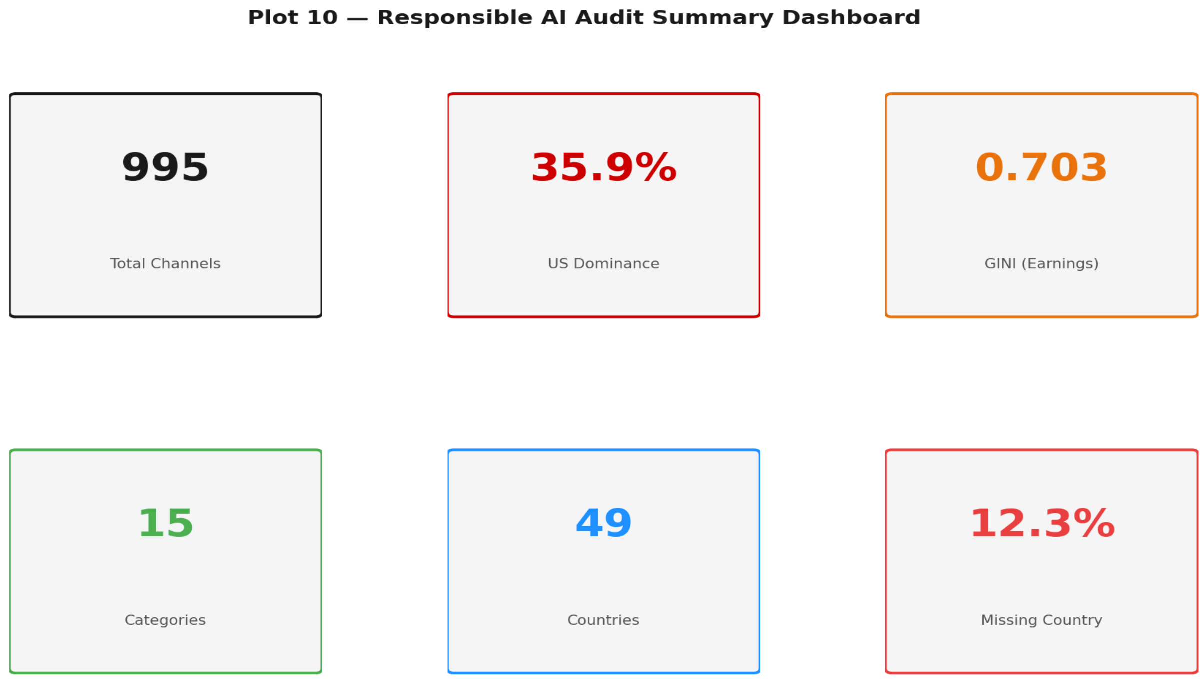

Figure 10: Key metrics summary dashboard from the full EDA audit.

■ What does this dashboard show?

This dashboard brings together the six most important numbers from our entire analysis into one view. Each card represents a key finding that has a direct ethical implication for Responsible AI.

■ Reading the Dashboard:

- **995 Total Channels** — A large, representative global sample.
- **US Dominance (35.9%)** — Geographic bias: 7× over-representation vs population share.
- **GINI Earnings (high)** — Severe earnings inequality among top creators.
- **15 Categories** — Wide content variety but unevenly rewarded by the algorithm.
- **47 Countries** — Global presence, but with heavy concentration in a few nations.
- **12.3% Missing Country** — Data gap that limits fairness auditing.

Conclusions & Responsible AI Recommendations

Bias Type	Metric Found	Severity	Recommended Fix
Popularity Bias	Subscribers drive rank	**HIGH**	Diversity re-ranking
Geographic Bias	US = 35.9% of top channels	**HIGH**	Country quota in Top-K
Earnings Inequality	GINI > 0.5 for earnings	**CRITICAL**	GINI penalty in ranking loss
Category Bias	Education < Entertainment earnings	**MEDIUM**	Category fairness weights
First-Mover Bias	Old channels dominate	**MEDIUM**	Recency boost for new creators
Explainability Gap	Rank drivers are opaque	**HIGH**	SHAP value reporting per channel
Data Privacy	12.3% missing country data	**MEDIUM**	Federated Learning for metadata

Overall Conclusion: YouTube's recommendation algorithm shows multiple forms of bias that are measurable, statistically significant, and ethically concerning. The system disproportionately rewards channels that are already popular, based in the United States, and focused on entertainment. Educational, science, and non-English-language creators face structural disadvantages. Implementing fairness-aware re-ranking, SHAP explainability, and geographic diversity constraints would make the platform significantly more equitable without sacrificing its overall performance.